Hotel Reservation & Flight Booking System

Short description.

Simple database that serves Hotel and Flight booking system has a well-designed structure that supports different types of analysis and reports, created using PostgreSQL.

Project Description and flow

Step 1 (Setting up tables that will be used in creating the database):

- Users (include all information about the system users and their roles)
- Airlines (Having data about the airlines)
- Hotels (include all essential data regarding hotels)
- Rooms (saves data regarding room details and pricing
- Flights (Having all essential data needed for upcoming flights)
- Seats (save all information needed for a certain flight seat)
- hotel_bookings (Relation table that allows users to make a hotel reservation)
- flight_bookings (Relation table that allows users to make a flight reservation)
- flight_booking_passengers (relation table that connects seats with flight booking)
- Payments (saves all payments transactions done on the system)
- Reviews (saves all reviews done on hotels and flights along the system)

Step 2 (Defining the attributes of these tables - First Schema):

- **users** (user_id(PK), first_name, last_name, email, phone_number, address, password, role, created_at)
- **airlines** (airline_id(PK), airline_name, country_code, phone, website, have_flights)
- **hotels** (hotel_id(PK), hotel_name, hotel_city, hotel_location, rating, description, fully_booked, created_at)
- **rooms** (room_id(PK), hotel_id(FK), room_type, price, is_available, created_at)
- **flights** (flight_id(PK), airline_id(FK), departure_city, arrival_city, departure_time, arrival_time, price, total_seats, available_seats, status, created_at)
- **seats** (seat_id(PK), flight_id(FK), seat_number, seat_class, is_available, created_at)
- **hotel_bookings** (booking_id(PK), user_id(FK), room_id(FK), check_in_date, check_out_date, number_of_guests, bill, booking_status, is_completed, created_at)
- **flight_bookings** (booking_id(PK), user_id(FK), number_of_passengers, bill, booking_status, is_completed, created_at)
- **flight_booking_passengers (**passenger_id(PK), flight_booking_id(FK), seat_id(FK), passenger_name, passenger_email, created_at)
- **payments** (payment_id(PK), hotel_booking_id(FK-UNIQUE), flight_booking_id(FK-UNIQUE), amount, payment_method, payment_status, payment_date)
- **reviews** (review_id(PK), user_id(FK), hotel_id(FK), flight_id(FK), rating, review_description, created_at)

Step 3 (Adding Relations Between Tables):

- rel1: users → hotel_bookings (1-M) One user can have multiple hotel bookings
- rel2: users → flight_bookings (1-M) One user can have multiple flight bookings
- rel3: hotels → rooms (1-M) One hotel has multiple rooms
- rel4: rooms → hotel_bookings (1-M) One room can be booked multiple times (different dates)
- rel5: airlines → flights (1-M) One airline operates multiple flights
- rel6: flights → seats (1-M) One flight has multiple seats

- rel7: flights → flight_bookings (1-M) One flight can have multiple bookings (through passengers)
- rel8: flight_bookings → flight_booking_passengers (1-M) One flight booking has multiple passengers
- rel9: seats → flight_booking_passengers (1-M) One seat is assigned to one passenger per booking
- rel10: hotel_bookings → payments (1-1) One hotel booking has exactly one payment
- rel11: flight_bookings → payments (1-1) One flight booking has exactly one payment
- rel12: users reviews (1-M) One user can write multiple reviews
- rel13: hotels reviews (1-M) One hotel can have multiple reviews
- rel14: flights reviews (1-M) One flight can have multiple reviews

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Tables Structures and Creation:

User table and its inserts

```
Users
CREATE TABLE users (
    user_id SERIAL PRIMARY KEY,
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    phone_number VARCHAR(20),
    address TEXT,
    password VARCHAR(255) NOT NULL,
    role VARCHAR(50) NOT NULL CHECK (role IN ('employee', 'admin', 'customer'))
);

 1. INSERT USERS
INSERT INTO users (first_name, last_name, email, phone_number, address, password, role)
VALUES
('Ahmed', 'Ali', 'ahmed.ali@email.com', '01001234567', '123 Cairo Street, Cairo', 'pass123',
   'customer'),
('Fatima', 'Mohamed', 'fatima.m@email.com', '01011234567', '456 Giza Avenue, Giza',
   'pass123', 'customer'),
('Hassan', 'Hassan', 'hassan.h@email.com', '01021234567', '789 Alex Road, Alexandria',
   'pass123', 'customer'),
('Sara', 'Ahmed', 'sara.ahmed@email.com', '01031234567', '321 Aswan Street, Aswan',
   'pass123', 'admin'),
```

```
('Omar', 'Ibrahim', 'omar.ibrahim@email.com', '01041234567', '654 Hurghada Lane, Hurghada',
   'pass123', 'employee'),
('Layla', 'Hassan', 'layla.hassan@email.com', '01051234567', '987 Sharm El-Sheikh, Sharm',
   'pass123', 'customer'),
('Karim', 'Khalil', 'karim.khalil@email.com', '01061234567', '111 Port Said, Port Said',
   'pass123', 'customer'),
('Noor', 'Nabil', 'noor.nabil@email.com', '01071234567', '222 Suez City, Suez', 'pass123',
   'customer');

select * from users;
```

Airlines table and its inserts

```
 Airlines
CREATE TABLE airlines (
    airline_id SERIAL PRIMARY KEY,
    airline_name VARCHAR(255) NOT NULL,
    country_code VARCHAR(2) NOT NULL,
    phone VARCHAR(15),
    website VARCHAR(255),
    have_flights BOOLEAN DEFAULT TRUE
);

 2. INSERT AIRLINES
INSERT INTO airlines (airline_name, country_code, phone, website, have_flights)
VALUES
('Egypt Air', 'EG', '02033460000', 'www.egyptair.com', TRUE),
('Air Cairo', 'EG', '02033460100', 'www.aircairo.com', TRUE),
('Emirates', 'AE', '04433310000', 'www.emirates.com', TRUE),
('Turkish Airlines', 'TR', '02166636363', 'www.turkishairlines.com', TRUE),
('Gulf Air', 'BH', '02243066000', 'www.gulfairco.com', TRUE);

INSERT INTO airlines (airline_name, country_code, phone, website, have_flights)
VALUES
('Nile Air', 'EG', '02033460000', 'www.nileair.com', FALSE)

select * from airlines;
```

Hotels table and its inserts

```
 Hotels
CREATE TABLE hotels (
    hotel_id SERIAL PRIMARY KEY,
    hotel_name VARCHAR(255) NOT NULL,
    hotel_city VARCHAR(100) NOT NULL,
    hotel_location VARCHAR(255),
    rating DECIMAL(2,1),
    description TEXT,
    fully_booked BOOLEAN DEFAULT FALSE
);


 3. INSERT HOTELS
INSERT INTO hotels (hotel_name, hotel_city, hotel_location, rating, description,
  fully_booked)
VALUES
('Nile Hilton', 'Cairo', 'Downtown Cairo, Nile Corniche', 5.0, 'Luxury 5-star hotel with
  stunning Nile views', FALSE),
('Pyramids Plaza', 'Giza', 'Giza Plateau, Near Pyramids', 4.5, '4-star hotel near the Great
  Pyramids', FALSE),
('Red Sea Resort', 'Hurghada', 'Hurghada Beach Front', 4.8, 'Beach resort with water sports
  facilities', FALSE),
('Sinai Paradise', 'Sharm El-Sheikh', 'Sharm Beach, Red Sea', 4.2, 'Family-friendly beach
  hotel', FALSE),
('Alexandria Grandeur', 'Alexandria', 'Alexandria Corniche', 4.0, 'Historic hotel with
  Mediterranean views', FALSE),
('Aswan Felucca', 'Aswan', 'Aswan Corniche, Nile River', 4.3, 'Traditional Nile-side hotel',
  TRUE);

select * from hotels;
```

Rooms table and its inserts

```
 Rooms
CREATE TABLE rooms (
    room_id SERIAL PRIMARY KEY,
    hotel_id INT NOT NULL,
    room_type VARCHAR(50) NOT NULL,   'single', 'double', 'suite'
    price DECIMAL(10,2) NOT NULL,
    is_available BOOLEAN DEFAULT TRUE,

    FOREIGN KEY (hotel_id) REFERENCES hotels(hotel_id),
    CHECK (price > 0)
);

 4. INSERT ROOMS (depends on hotels)
INSERT INTO rooms (hotel_id, room_type, price, is_available)
VALUES
(1, 'single', 150.00, TRUE),
(1, 'double', 250.00, TRUE),
(1, 'suite', 450.00, FALSE),
(2, 'single', 100.00, TRUE),
(2, 'double', 180.00, TRUE),
(3, 'double', 200.00, TRUE),
(3, 'suite', 380.00, TRUE),
(4, 'single', 80.00, FALSE),
(4, 'double', 160.00, TRUE),
(5, 'double', 140.00, TRUE);

INSERT INTO rooms (hotel_id, room_type, price, is_available)
VALUES (1, 'single', 200.00, TRUE);

INSERT INTO rooms (hotel_id, room_type, price, is_available) VALUES
(1, 'single', 180.00, TRUE),
(1, 'double', 260.00, TRUE),
(1, 'suite', 500.00, TRUE),
(1, 'single', 170.00, TRUE);

select * from rooms;
```

Flights table and its inserts

```
 Flights
CREATE TABLE flights (
    flight_id SERIAL PRIMARY KEY,
    airline_id INT NOT NULL,
    departure_city VARCHAR(100) NOT NULL,
    arrival_city VARCHAR(100) NOT NULL,
    departure_time TIMESTAMP NOT NULL,
    arrival_time TIMESTAMP NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    total_seats INT NOT NULL,
    available_seats INT NOT NULL,
    status VARCHAR(50) DEFAULT 'scheduled',

    FOREIGN KEY (airline_id) REFERENCES airlines(airline_id),
    CHECK (available_seats >= 0 AND available_seats <= total_seats),
    CHECK (departure_time < arrival_time),
    CHECK (price > 0)
);
 5. INSERT FLIGHTS (depends on airlines)
INSERT INTO flights (airline_id, departure_city, arrival_city, departure_time, arrival_time,
  price, total_seats, available_seats, status)
VALUES
(1, 'Cairo', 'London', '2026-04-10 08:00:00', '2026-04-10 14:30:00', 500.00, 180, 45,
  'scheduled'),
(1, 'Cairo', 'Paris', '2026-04-11 10:00:00', '2026-04-11 16:00:00', 480.00, 180, 80,
  'scheduled'),
(2, 'Cairo', 'Dubai', '2026-04-12 06:00:00', '2026-04-12 10:30:00', 350.00, 150, 30,
  'scheduled'),
(3, 'Cairo', 'Abu Dhabi', '2026-04-13 09:00:00', '2026-04-13 11:00:00', 320.00, 200, 120,
  'scheduled'),
(4, 'Cairo', 'Istanbul', '2026-04-14 07:00:00', '2026-04-14 12:00:00', 420.00, 160, 60,
  'scheduled'),
(5, 'Cairo', 'Doha', '2026-04-15 11:00:00', '2026-04-15 13:30:00', 280.00, 140, 85,
  'scheduled'),
(1, 'Alexandria', 'London', '2026-04-16 09:30:00', '2026-04-16 16:00:00', 520.00, 180, 50,
  'scheduled'),
(2, 'Hurghada', 'Frankfurt', '2026-04-17 08:00:00', '2026-04-17 15:00:00', 450.00, 150, 40,
  'scheduled');
INSERT INTO flights (airline_id, departure_city, arrival_city, departure_time, arrival_time,
  price, total_seats, available_seats)
VALUES
```

```
(1, 'Cairo', 'London', '2026-04-18 08:00:00', '2026-04-18 14:00:00', 450.00, 180, 100),
(2, 'Cairo', 'London', '2026-04-19 09:00:00', '2026-04-19 15:00:00', 550.00, 150, 20);

select * from flights;
```

Seats table and its inserts

```
 Seats
CREATE TABLE seats (
    seat_id SERIAL PRIMARY KEY,
    flight_id INT NOT NULL,
    seat_number VARCHAR(10) NOT NULL,
    seat_class VARCHAR(20) NOT NULL,
    is_available BOOLEAN DEFAULT TRUE,

    FOREIGN KEY (flight_id) REFERENCES flights(flight_id),
    UNIQUE(flight_id, seat_number)
);

 6. INSERT SEATS (depends on flights)
INSERT INTO seats (flight_id, seat_number, seat_class, is_available)
VALUES
 Flight 1 (Cairo-London)
(1, '1A', 'economy', TRUE),
(1, '1B', 'economy', TRUE),
(1, '1C', 'economy', FALSE),
(1, '2A', 'business', TRUE),
(1, '2B', 'business', FALSE),
 Flight 2 (Cairo-Paris)
(2, '1A', 'economy', TRUE),
(2, '1B', 'economy', TRUE),
(2, '2A', 'business', TRUE),
(2, '2B', 'business', TRUE),
 Flight 3 (Cairo-Dubai)
(3, '1A', 'economy', TRUE),
(3, '1B', 'economy', FALSE),
(3, '2A', 'business', TRUE),
 Flight 4 (Cairo-Abu Dhabi)
(4, '1A', 'economy', TRUE),
(4, '1B', 'economy', TRUE),
(4, '2A', 'business', FALSE),
 Flight 5 (Cairo-Istanbul)
```

```
(5, '1A', 'economy', TRUE),
(5, '1B', 'economy', TRUE),
(5, '2A', 'business', TRUE),
 Flight 6 (Cairo-Doha)
(6, '1A', 'economy', TRUE),
(6, '1B', 'economy', TRUE),
 Flight 7 (Alexandria-London)
(7, '1A', 'economy', TRUE),
(7, '2A', 'business', TRUE),
 Flight 8 (Hurghada-Frankfurt)
(8, '1A', 'economy', TRUE),
(8, '2A', 'business', TRUE);

INSERT INTO seats (flight_id, seat_number, seat_class, is_available)
VALUES
(9, '1A', 'economy', TRUE),
(9, '1B', 'economy', FALSE),
(10, '1A', 'economy', TRUE),
(10, '2A', 'business', TRUE);

select * from seats;
```

Hotel_b ookings table and its inserts

```
 Hotel Bookings
CREATE TABLE hotel_bookings (
    booking_id SERIAL PRIMARY KEY,
    user_id INT NOT NULL,
    room_id INT NOT NULL,
    check_in_date DATE NOT NULL,
    check_out_date DATE NOT NULL,
    number_of_guests INT NOT NULL,
    bill DECIMAL(10,2) NOT NULL,
    booking_status VARCHAR(50) DEFAULT 'pending',
    is_completed BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (user_id) REFERENCES users(user_id),
    FOREIGN KEY (room_id) REFERENCES rooms(room_id),
    CHECK (check_in_date < check_out_date),
    CHECK (number_of_guests > 0),
```

```
    CHECK (bill >= 0)

);

 7. INSERT HOTEL BOOKINGS
INSERT INTO hotel_bookings (user_id, room_id, check_in_date, check_out_date,
  number_of_guests, bill, booking_status, is_completed)
VALUES
(1, 1, '2026-04-15', '2026-04-17', 1, 300.00, 'confirmed', FALSE),
(2, 2, '2026-04-18', '2026-04-21', 2, 750.00, 'confirmed', FALSE),
(3, 4, '2026-04-10', '2026-04-12', 1, 200.00, 'pending', FALSE),
(4, 6, '2026-04-20', '2026-04-23', 2, 600.00, 'confirmed', FALSE),
(5, 7, '2026-04-12', '2026-04-14', 2, 760.00, 'confirmed', FALSE),
(6, 9, '2026-04-25', '2026-04-27', 2, 320.00, 'pending', FALSE),
(7, 10, '2026-04-18', '2026-04-20', 1, 140.00, 'cancelled', FALSE),
(8, 2, '2026-04-22', '2026-04-25', 2, 750.00, 'confirmed', FALSE);

INSERT INTO hotel_bookings (user_id, room_id, check_in_date, check_out_date,
  number_of_guests, bill, booking_status)
VALUES
(1, 11, '2026-04-10', '2026-04-12', 1, 360.00, 'confirmed'),
(2, 12, '2026-04-11', '2026-04-13', 2, 520.00, 'cancelled'),
(3, 13, '2026-04-12', '2026-04-15', 2, 1000.00, 'confirmed');

select * from hotel_bookings;
```

flight_bookings table and its inserts

```
Flight Booking
CREATE TABLE flight_bookings (
    booking_id SERIAL PRIMARY KEY,
    user_id INT NOT NULL,
    number_of_passengers INT NOT NULL,
    bill DECIMAL(10,2) NOT NULL,
    booking_status VARCHAR(50) DEFAULT 'pending',
    is_completed BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (user_id) REFERENCES users(user_id),
    CHECK (number_of_passengers > 0),
    CHECK (bill >= 0)
```

```sql
);


 8. INSERT FLIGHT BOOKINGS
INSERT INTO flight_bookings (user_id, number_of_passengers, bill, booking_status,
  is_completed)
VALUES
(1, 1, 500.00, 'confirmed', FALSE),
(2, 2, 1000.00, 'confirmed', FALSE),
(3, 1, 350.00, 'pending', FALSE),
(4, 3, 960.00, 'confirmed', FALSE),
(5, 2, 840.00, 'confirmed', FALSE),
(6, 1, 280.00, 'pending', FALSE),
(7, 2, 1040.00, 'confirmed', FALSE),
(8, 1, 450.00, 'cancelled', FALSE);

select * from flight_bookings;
```

Payments table and its inserts

```sql


 Payments
CREATE TABLE payments (
    payment_id SERIAL PRIMARY KEY,
    hotel_booking_id INT,
    flight_booking_id INT,
    amount DECIMAL(10,2) NOT NULL,
    payment_method VARCHAR(50),
    payment_status VARCHAR(50) DEFAULT 'pending',
    payment_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (hotel_booking_id) REFERENCES hotel_bookings(booking_id),
    FOREIGN KEY (flight_booking_id) REFERENCES flight_bookings(booking_id),
    CHECK (hotel_booking_id IS NOT NULL OR flight_booking_id IS NOT NULL),
    CHECK (amount > 0)
);


 10. INSERT PAYMENTS
INSERT INTO payments (hotel_booking_id, flight_booking_id, amount, payment_method,
  payment_status, payment_date)
VALUES
```

```sql
(1, NULL, 300.00, 'credit_card', 'completed', '2026-04-14 10:30:00'),
(2, NULL, 750.00, 'debit_card', 'completed', '2026-04-17 14:15:00'),
(3, NULL, 200.00, 'online_transfer', 'pending', '2026-04-09 16:45:00'),
(4, NULL, 600.00, 'credit_card', 'completed', '2026-04-19 11:20:00'),
(5, NULL, 760.00, 'debit_card', 'completed', '2026-04-11 09:00:00'),
(6, NULL, 320.00, 'credit_card', 'pending', '2026-04-24 15:30:00'),
(7, NULL, 140.00, 'credit_card', 'cancelled', '2026-04-17 12:00:00'),
(8, NULL, 750.00, 'online_transfer', 'completed', '2026-04-21 13:45:00'),
(NULL, 1, 500.00, 'credit_card', 'completed', '2026-04-09 08:30:00'),
(NULL, 2, 1000.00, 'debit_card', 'completed', '2026-04-10 10:15:00'),
(NULL, 3, 350.00, 'credit_card', 'pending', '2026-04-11 14:20:00'),
(NULL, 4, 960.00, 'online_transfer', 'completed', '2026-04-12 16:45:00'),
(NULL, 5, 840.00, 'credit_card', 'completed', '2026-04-13 09:30:00'),
(NULL, 6, 280.00, 'debit_card', 'pending', '2026-04-14 11:00:00'),
(NULL, 7, 1040.00, 'credit_card', 'completed', '2026-04-15 12:15:00'),
(NULL, 8, 450.00, 'online_transfer', 'cancelled', '2026-04-16 14:30:00');

INSERT INTO payments (hotel_booking_id, amount, payment_method, payment_status)
VALUES
(9, 360.00, 'credit_card', 'completed'),
(10, 520.00, 'debit_card', 'cancelled'),
(11, 1000.00, 'credit_card', 'completed');

select * from payments;
```

Reviews table and its inserts

```sql
 Reviews
CREATE TABLE reviews (
    review_id SERIAL PRIMARY KEY,
    user_id INT NOT NULL,
    hotel_id INT,
    flight_id INT,
    rating INT NOT NULL CHECK (rating BETWEEN 1 AND 5),
    review_description TEXT,

    FOREIGN KEY (user_id) REFERENCES users(user_id),
    FOREIGN KEY (hotel_id) REFERENCES hotels(hotel_id),
    FOREIGN KEY (flight_id) REFERENCES flights(flight_id),
    CHECK (hotel_id IS NOT NULL OR flight_id IS NOT NULL)
```

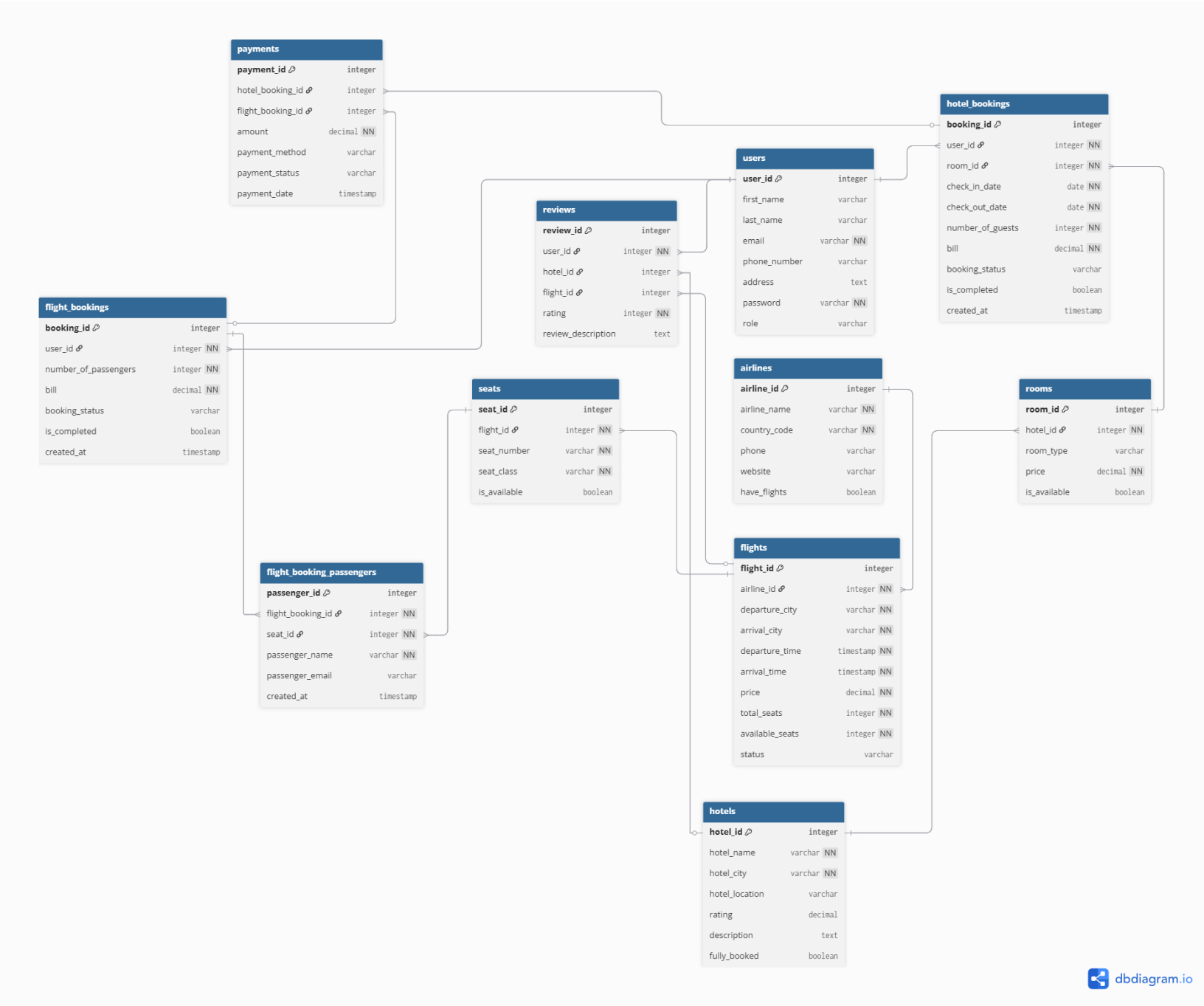
```sql
);

 11. INSERT REVIEWS
INSERT INTO reviews (user_id, hotel_id, flight_id, rating, review_description)
VALUES
(1, 1, NULL, 5, 'Excellent service and great location! The staff was very friendly.'),
(2, 2, NULL, 4, 'Good hotel near the pyramids. Rooms were clean and comfortable.'),
(3, 3, NULL, 5, 'Amazing beach resort! Perfect for relaxation and water sports.'),
(4, 4, NULL, 3, 'Nice hotel but rooms need renovation. Food was average.'),
(5, 5, NULL, 4, 'Historic hotel with character. Mediterranean views are stunning.'),
(1, NULL, 1, 5, 'Great flight experience. On-time arrival and friendly crew.'),
(2, NULL, 2, 4, 'Good service but seats were a bit cramped.'),
(3, NULL, 3, 5, 'Excellent service and smooth flight. Highly recommend!'),
(4, NULL, 4, 4, 'Professional crew and comfortable seating.'),
(6, NULL, 5, 4, 'Good value for money. Flight was smooth.'),
(7, 2, NULL, 3, 'Decent hotel. Could use better Wi-Fi.'),
(8, 1, NULL, 5, 'Loved every moment at Nile Hilton! Perfect for my stay.');

select * from reviews;
```

flight_booking_passengers table and its inserts

```sql
 Flight Booking Passengers
CREATE TABLE flight_booking_passengers (
    passenger_id SERIAL PRIMARY KEY,
    flight_booking_id INT NOT NULL,
    seat_id INT NOT NULL,
    passenger_name VARCHAR(100) NOT NULL,
    passenger_email VARCHAR(100),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (flight_booking_id) REFERENCES flight_bookings(booking_id),
    FOREIGN KEY (seat_id) REFERENCES seats(seat_id),
    UNIQUE(seat_id)
);

 9. INSERT FLIGHT BOOKING PASSENGERS
INSERT INTO flight_booking_passengers (flight_booking_id, seat_id, passenger_name,
  passenger_email)
VALUES
(1, 1, 'Ahmed Ali', 'ahmed.ali@email.com'),
(2, 2, 'Fatima Mohamed', 'fatima.m@email.com'),
```

```
(2, 4, 'Hassan Hassan', 'hassan.h@email.com'),
(3, 10, 'Sara Ahmed', 'sara.ahmed@email.com'),
(4, 13, 'Omar Ibrahim', 'omar.ibrahim@email.com'),
(4, 14, 'Layla Hassan', 'layla.hassan@email.com'),
(4, 15, 'Karim Khalil', 'karim.khalil@email.com'),
(5, 16, 'Noor Nabil', 'noor.nabil@email.com'),
(5, 17, 'Ahmed Ali', 'ahmed.ali@email.com'),
(6, 19, 'Fatima Mohamed', 'fatima.m@email.com'),
(7, 21, 'Hassan Hassan', 'hassan.h@email.com'),
(7, 22, 'Sara Ahmed', 'sara.ahmed@email.com'),
(8, 24, 'Omar Ibrahim', 'omar.ibrahim@email.com');

INSERT INTO flight_booking_passengers (flight_booking_id, seat_id, passenger_name)
VALUES
(1, 25, 'New Passenger');

select * from flight_booking_passengers;
```

System Entity Relation Diagram:



Project Sample Queries

show hotels by location (City = 'Cairo')

```
SELECT hotel_id, hotel_name, hotel_city, hotel_location, rating, description
FROM hotels
WHERE hotel_city = 'Cairo'
  AND NOT fully_booked
ORDER BY rating DESC, hotel_name;
```

Output:

	hotel_name character varying (255)	hotel_city character varying (100)	hotel_location character varying (255)	rating numeric (2,1)	description text
1	Nile Hilton	Cairo	Downtown Cairo, Nile Cornic...	5.0	Luxury 5-star hotel with stunning Nile vie...

show all the available rooms in the hotel in previous selection

```
SELECT  h.hotel_name, h.hotel_city, r.room_id, r.room_type, r.price AS price_per_night
FROM hotels h
JOIN rooms r ON h.hotel_id = r.hotel_id
WHERE h.hotel_city = 'Cairo' AND NOT h.fully_booked AND r.is_available = TRUE
AND NOT EXISTS (SELECT 1 FROM hotel_bookings hb WHERE hb.room_id = r.room_id AND
hb.booking_status != 'cancelled')
ORDER BY h.rating DESC, h.hotel_name;
```

Output:

	hotel_name character varying (255)	hotel_city character varying (100)	room_id integer	room_type character varying (50)	price_per_night numeric (10,2)
1	Nile Hilton	Cairo	12	single	162.00
2	Nile Hilton	Cairo	15	single	153.00
3	Nile Hilton	Cairo	14	suite	375.00

if customers want to go on a summer vacation in Hurghada and need to find the rooms available on specific date

Hurghada, April 15-17

```
SELECT  h.hotel_name, h.hotel_city, r.room_id, r.room_type, r.price as Price_Per_night
FROM hotels h JOIN rooms r ON h.hotel_id = r.hotel_id
LEFT JOIN hotel_bookings hb ON r.room_id = hb.room_id AND hb.booking_status != 'cancelled'
AND NOT (hb.check_out_date <= '2026-04-15' OR hb.check_in_date >= '2026-04-17')
WHERE h.hotel_city = 'Hurghada'
AND r.is_available = TRUE
AND hb.booking_id IS NULL
```

Output:

	hotel_name character varying (255)	hotel_city character varying (100)	room_id integer	room_type character varying (50)	price_per_night numeric (10,2)
1	Red Sea Resort	Hurghada	6	double	200.00
2	Red Sea Resort	Hurghada	7	suite	380.00

Hurghada, April 20-23

```
SELECT  h.hotel_name, h.hotel_city, r.room_id, r.room_type, r.price as Price_Per_night
FROM hotels h JOIN rooms r ON h.hotel_id = r.hotel_id
LEFT JOIN hotel_bookings hb ON r.room_id = hb.room_id AND hb.booking_status != 'cancelled'
AND NOT (hb.check_out_date <= '2026-04-20' OR hb.check_in_date >= '2026-04-23')
WHERE h.hotel_city = 'Hurghada'
AND r.is_available = TRUE
AND hb.booking_id IS NULL
```

Output:

	hotel_name character varying (255)	hotel_city character varying (100)	room_id integer	room_type character varying (50)	price_per_night numeric (10,2)
1	Red Sea Resort	Hurghada	7	suite	380.00

show the best hotels in a certain city that are not fully booked

```
SELECT hotel_name, hotel_city, rating
FROM hotels
WHERE hotel_city = 'Cairo' AND NOT fully_booked
ORDER BY rating DESC;
```

Output:

	hotel_name character varying (255)	hotel_city character varying (100)	rating numeric (2,1)
1	Nile Hilton	Cairo	5.0

show total revenue per hotel like which hotel make more money ()

```
SELECT h.hotel_name, SUM(p.amount) AS revenue
FROM hotels h
JOIN rooms r ON h.hotel_id = r.hotel_id
JOIN hotel_bookings hb ON r.room_id = hb.room_id
JOIN payments p ON hb.booking_id = p.hotel_booking_id
WHERE p.payment_status = 'completed'
GROUP BY h.hotel_name
ORDER BY revenue DESC;
```

Output:

	hotel_name character varying (255)	revenue numeric
1	Nile Hilton	3160.00
2	Red Sea Resort	1360.00

show available flights from cairo to london

```
SELECT f.flight_id, a.airline_name, f.departure_city, f.arrival_city, f.price
FROM flights f
JOIN airlines a ON f.airline_id = a.airline_id
WHERE f.departure_city = 'Cairo' AND f.arrival_city = 'London' AND f.available_seats > 0
ORDER BY f.price ;
```

Output:

	flight_id integer	airline_name character varying (255)	departure_city character varying (100)	arrival_city character varying (100)	price numeric (10,2)
1	9	Egypt Air	Cairo	London	450.00
2	1	Egypt Air	Cairo	London	500.00
3	10	Air Cairo	Cairo	London	550.00

show all booking history of flights and hotels (booking report)

```
SELECT u.first_name, 'Hotel' AS type, hb.booking_id, hb.bill, hb.booking_status
FROM users u
JOIN hotel_bookings hb ON u.user_id = hb.user_id
UNION ALL
SELECT u.first_name, 'Flight' AS type, fb.booking_id, fb.bill, fb.booking_status
FROM users u
JOIN flight_bookings fb ON u.user_id = fb.user_id;
```

Output:

	first_name character varying (100)	type text	booking_id integer	bill numeric (10,2)	booking_status character varying (50)
1	Ahmed	Hotel	1	300.00	confirmed
2	Fatima	Hotel	2	750.00	confirmed
3	Hassan	Hotel	3	200.00	pending
4	Sara	Hotel	4	600.00	confirmed
5	Omar	Hotel	5	760.00	confirmed
6	Layla	Hotel	6	320.00	pending
7	Karim	Hotel	7	140.00	cancelled
8	Noor	Hotel	8	750.00	confirmed
9	Ahmed	Hotel	9	360.00	confirmed
10	Fatima	Hotel	10	520.00	cancelled
11	Hassan	Hotel	11	1000.00	confirmed
12	Ahmed	Hotel	12	400.00	confirmed
13	Ahmed	Flight	1	500.00	confirmed
14	Fatima	Flight	2	1000.00	confirmed
15	Hassan	Flight	3	350.00	pending
16	Sara	Flight	4	960.00	confirmed
17	Omar	Flight	5	840.00	confirmed
18	Layla	Flight	6	280.00	pending
19	Karim	Flight	7	1040.00	confirmed
20	Noor	Flight	8	450.00	cancelled

show all users who have booked hotels

SELECT u.user_id, u.first_name, u.last_name, 'Hotel' AS type, hb.booking_id, hb.bill, hb.booking_status FROM users u JOIN hotel_bookings hb ON u.user_id = hb.user_id;

Output:

	user_id integer	first_name character varying (100)	last_name character varying (100)	type text	booking_id integer	bill numeric (10,2)	booking_status character varying (50)
1	1	Ahmed	Ali	Hotel	1	300.00	confirmed
2	2	Fatima	Mohamed	Hotel	2	750.00	confirmed
3	3	Hassan	Hassan	Hotel	3	200.00	pending
4	4	Sara	Ahmed	Hotel	4	600.00	confirmed
5	5	Omar	Ibrahim	Hotel	5	760.00	confirmed
6	6	Layla	Hassan	Hotel	6	320.00	pending
7	7	Karim	Khalil	Hotel	7	140.00	cancelled
8	8	Noor	Nabil	Hotel	8	750.00	confirmed
9	1	Ahmed	Ali	Hotel	9	360.00	confirmed
10	2	Fatima	Mohamed	Hotel	10	520.00	cancelled
11	3	Hassan	Hassan	Hotel	11	1000.00	confirmed
12	1	Ahmed	Ali	Hotel	12	400.00	confirmed

show all users who have booked flights

SELECT u.user_id, u.first_name, u.last_name, 'Flight' AS type, fb.booking_id, fb.bill, fb.booking_status FROM users u JOIN flight_bookings fb ON u.user_id = fb.user_id;

Output:

	user_id integer	first_name character varying (100)	last_name character varying (100)	type text	booking_id integer	bill numeric (10,2)	booking_status character varying (50)
1	1	Ahmed	Ali	Flight	1	500.00	confirmed
2	2	Fatima	Mohamed	Flight	2	1000.00	confirmed
3	3	Hassan	Hassan	Flight	3	350.00	pending
4	4	Sara	Ahmed	Flight	4	960.00	confirmed
5	5	Omar	Ibrahim	Flight	5	840.00	confirmed
6	6	Layla	Hassan	Flight	6	280.00	pending
7	7	Karim	Khalil	Flight	7	1040.00	confirmed
8	8	Noor	Nabil	Flight	8	450.00	cancelled

show all users who have booked both hotels and flights

```
SELECT u.user_id, u.first_name, u.last_name
FROM users u
JOIN hotel_bookings hb ON u.user_id = hb.user_id
INTERSECT
SELECT u.user_id, u.first_name, u.last_name
FROM users u
JOIN flight_bookings fb ON u.user_id = fb.user_id;
```

Output:

	user_id integer	first_name character varying (100)	last_name character varying (100)
1	8	Noor	Nabil
2	6	Layla	Hassan
3	1	Ahmed	Ali
4	2	Fatima	Mohamed
5	5	Omar	Ibrahim
6	4	Sara	Ahmed
7	3	Hassan	Hassan
8	7	Karim	Khalil

calculate all the bills or money that we need to collect (Collect Money Report)

```
SELECT hb.booking_id,(hb.check_out_date - hb.check_in_date) * r.price AS calculated_bill
FROM hotel_bookings hb
JOIN rooms r ON hb.room_id = r.room_id;
```

Output:

	booking_id [PK] integer	calculated_bill numeric
1	1	270.00
2	2	637.50
3	3	200.00
4	4	600.00
5	5	760.00
6	6	320.00
7	7	280.00
8	8	637.50
9	9	360.00
10	10	324.00
11	11	663.00
12	12	360.00

show all hotels that have no bookings

```
SELECT h.hotel_name ,h.hotel_city FROM hotels h
WHERE NOT EXISTS (
    SELECT 1 FROM rooms r
    JOIN hotel_bookings hb ON r.room_id = hb.room_id
    WHERE r.hotel_id = h.hotel_id
);
```

Output:

	hotel_name character varying (255)	hotel_city character varying (100)
1	Aswan Felucca	Aswan

show the most expensive luxurious rooms available in our hotels

```
SELECT h.hotel_name, r.room_id, r.room_type, r.price
FROM rooms r
JOIN hotels h ON r.hotel_id = h.hotel_id
WHERE r.is_available = TRUE AND r.price = (
    SELECT MAX(price)
    FROM rooms r2
    WHERE r2.hotel_id = r.hotel_id)
ORDER BY r.price DESC;
```

Output:

	hotel_name character varying (255)	room_id integer	room_type character varying (50)	price numeric (10,2)
1	Red Sea Resort	7	suite	380.00
2	Nile Hilton	14	suite	375.00
3	Pyramids Plaza	5	double	180.00
4	Sinai Paradise	9	double	160.00
5	Alexandria Grandeur	10	double	140.00

user book a room in an hotel

```
BEGIN;
INSERT INTO hotel_bookings (user_id, room_id, check_in_date, check_out_date, number_of_guests,
bill, booking_status)
VALUES (1, 11, '2026-05-01', '2026-05-03', 2, 400, 'confirmed');
UPDATE rooms SET is_available = FALSE WHERE room_id = 11;
COMMIT;
```

users who booked a flight for his family

```
SELECT u.first_name, fb.booking_id, COUNT(p.passenger_id) AS passengers,fb.booking_status
FROM flight_bookings fb
JOIN flight_booking_passengers p ON fb.booking_id = p.flight_booking_id
JOIN users u ON u.user_id= fb.user_id
GROUP BY u.first_name,fb.booking_id;
```

Output:

	first_name character varying (100)	booking_id integer	passengers bigint	booking_status character varying (50)
1	Ahmed	1	2	confirmed
2	Omar	5	2	confirmed
3	Sara	4	3	confirmed
4	Karim	7	2	confirmed
5	Fatima	2	2	confirmed
6	Hassan	3	1	pending
7	Noor	8	1	cancelled
8	Layla	6	1	pending

show all flights that have no passengers (Empty Flight Report)

```
SELECT a.airline_name, f.flight_id
FROM flights f
JOIN airlines a ON a.airline_id=f.airline_id
WHERE NOT EXISTS (
    SELECT 1 FROM seats s
    JOIN flight_booking_passengers p ON s.seat_id = p.seat_id
    WHERE s.flight_id = f.flight_id
);
```

Output:

	airline_name character varying (255)	flight_id integer
1	Air Cairo	10
2	Egypt Air	2

compare revenue of both hotels and flights (Decision making)

```
SELECT 'Hotel' AS Type, SUM(amount)
FROM payments
WHERE hotel_booking_id IS NOT NULL
UNION ALL
SELECT 'Flight', SUM(amount)
FROM payments
WHERE flight_booking_id IS NOT NULL;
```

Output:

	type text	sum numeric
1	Hotel	5700.00
2	Flight	5420.00

show reviews of hotels

```
SELECT h.hotel_name, COUNT(r.review_id) as num_of_reviews, ROUND(AVG(r.rating),2) as
Average_Rate
FROM hotels h
JOIN reviews r ON h.hotel_id = r.hotel_id
GROUP BY h.hotel_name
ORDER BY Average_Rate;
```

Output:

	hotel_name character varying (255)	num_of_reviews bigint	average_rate numeric
1	Sinai Paradise	1	3.00
2	Pyramids Plaza	2	3.50
3	Alexandria Grandeur	1	4.00
4	Nile Hilton	2	5.00
5	Red Sea Resort	1	5.00

show all payment status for admin (Financial Report)

```
SELECT p.payment_id, u.first_name || ' ' || u.last_name AS customer_name, p.amount,
p.payment_method,
p.payment_status, p.payment_date, CASE WHEN p.payment_status = 'completed' THEN p.amount
ELSE 0 END AS revenue
FROM payments p
LEFT JOIN hotel_bookings hb ON p.hotel_booking_id = hb.booking_id
LEFT JOIN flight_bookings fb ON p.flight_booking_id = fb.booking_id
LEFT JOIN users u ON COALESCE(hb.user_id, fb.user_id) = u.user_id
ORDER BY p.payment_date DESC;
```

Output:

	payment_id [PK] integer	customer_name text	amount numeric (10,2)	payment_method character varying (50)	payment_status character varying (50)	payment_date timestamp without time zone	revenue numeric
1	6	Layla Hassan	320.00	credit_card	pending	2026-04-24 15:30:00	0
2	8	Noor Nabil	750.00	online_transfer	completed	2026-04-21 13:45:00	750.00
3	4	Sara Ahmed	600.00	credit_card	completed	2026-04-19 11:20:00	600.00
4	2	Fatima Mohamed	750.00	debit_card	completed	2026-04-17 14:15:00	750.00
5	7	Karim Khalil	140.00	credit_card	cancelled	2026-04-17 12:00:00	0
6	16	Noor Nabil	450.00	online_transfer	cancelled	2026-04-16 14:30:00	0
7	15	Karim Khalil	1040.00	credit_card	completed	2026-04-15 12:15:00	1040.00
8	14	Layla Hassan	280.00	debit_card	pending	2026-04-14 11:00:00	0
9	1	Ahmed Ali	300.00	credit_card	completed	2026-04-14 10:30:00	300.00
10	13	Omar Ibrahim	840.00	credit_card	completed	2026-04-13 09:30:00	840.00
11	12	Sara Ahmed	960.00	online_transfer	completed	2026-04-12 16:45:00	960.00
12	11	Hassan Hassan	350.00	credit_card	pending	2026-04-11 14:20:00	0
13	5	Omar Ibrahim	760.00	debit_card	completed	2026-04-11 09:00:00	760.00
14	10	Fatima Mohamed	1000.00	debit_card	completed	2026-04-10 10:15:00	1000.00
15	3	Hassan Hassan	200.00	online_transfer	pending	2026-04-09 16:45:00	0
16	9	Ahmed Ali	500.00	credit_card	completed	2026-04-09 08:30:00	500.00
17	17	Ahmed Ali	360.00	credit_card	completed	2026-03-25 16:39:30.134012	360.00
18	18	Fatima Mohamed	520.00	debit_card	cancelled	2026-03-25 16:39:30.134012	0
19	19	Hassan Hassan	1000.00	credit_card	completed	2026-03-25 16:39:30.134012	1000.00

COMPLETE FLIGHT BOOKING

-

```
BEGIN TRANSACTION;

 --creation of flight booking for 3 passengers
INSERT INTO flight_bookings (user_id, number_of_passengers, bill, booking_status)
VALUES (2, 3, 1500.00, 'pending');

 --Assign seat 1 to passenger 1
INSERT INTO flight_booking_passengers (flight_booking_id, seat_id, passenger_name,
passenger_email)
VALUES (1, 1, 'Ahmed Ali', 'ahmed@email.com');

 --Assign seat 2 to passenger 2
INSERT INTO flight_booking_passengers (flight_booking_id, seat_id, passenger_name,
passenger_email)
VALUES (1, 2, 'Fatima Mohamed', 'fatima@email.com');

-- Assign seat 3 to passenger 3
INSERT INTO flight_booking_passengers (flight_booking_id, seat_id, passenger_name,
passenger_email)
VALUES (1, 4, 'Hassan Hassan', 'hassan@email.com');

 --Record payment ONLY if all passengers assigned
INSERT INTO payments (flight_booking_id, amount, payment_method, payment_status)
VALUES (1, 1500.00, 'credit_card', 'completed');

-- Update flight available seats
UPDATE flights
SET available_seats = available_seats - 3
WHERE flight_id = 1;

 --Update seats as booked
UPDATE seats
SET is_available = FALSE
WHERE seat_id IN (1, 2, 4);

COMMIT;
ROLLBACK;
```

update a room someone just checked in

```
UPDATE hotel_bookings SET booking_status = 'checked_in', is_completed = FALSE
WHERE booking_id = 1 AND check_in_date = CURRENT_DATE
AND booking_status = 'confirmed';
```

apply some offers on rooms prices

```
UPDATE rooms SET price = CASE
WHEN room_type = 'suite' THEN price * 0.75   25% offer for suites
WHEN room_type = 'double' THEN price * 0.85  15% offer for doubles
WHEN room_type = 'single' THEN price * 0.90  10% offer for singles
END
WHERE hotel_id = 1
AND room_type IN ('single', 'double', 'suite');
```